Yunhui Song

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Research Interests

I design and study interactive systems that connect the digital and physical worlds through spatiotemporal AR/VR interfaces and wearable technologies. By combining sensing and feedback, my work aims to augment human physical capabilities and foster more immersive and engaging interactions.

Education

Aarhus University, Denmark *Sept. 2024 – Jul. 2026 (expected)*
M.Sc. in Computer Science

Chengdu University of Technology, China *Sept. 2017 – Jun. 2021*
B.Eng. in Computer Science and Technology

Academic Experience

Human-Centered Computing Section, Aarhus University *Jan. 2026 – Present*
Research Intern in 4D Gaussian Splatting for Immersive VR, Supervised by ***Prof. Stefanie Zollmann***
Human-Centered Computing Section, Aarhus University *Nov. 2024 – Sept. 2025*
Research Intern in Intraoral User Interfaces, Supervised by ***Prof. Michael Wessely***

Master Project

Scene360: 360° Panoramic Scene Reconstruction and 4D Gaussian Splatting-Based Dynamic Object Composition
Supervised by Prof. Stefanie Zollmann *Jan. 2026 – Present*
Developed a pipeline for 360° panoramic scene reconstruction and dynamic object composition, covering static scene modeling, 4D Gaussian-based dynamic reconstruction, and scene integration. Built a rendering and visualization framework for immersive exploration of reconstructed scenes.

Peer-Reviewed Publications

*=EQUAL CONTRIBUTION

TeethMark: Enabling Tongue–Tooth Interaction with Electrotactile Feedback using an Intraoral Interface.
Yijing Jiang, **Yunhui Song**, Haiyang Xu, Maria Perdices Segui, Himani Deshpande, and Michael Wessely. *(Under Review at UIST2026)*.

EchoMind: Augmenting Recorded Presentations with Context-Grounded Conversational Agents and Collective Viewer Signals.
Yunhui Song,* Haiyang Xu,* Black Sun and Niklas Elmqvist. *(Under Review at NordiCHI2026)*.

Non-Peer-Reviewed Publications

Demonstration of BIOral: Fabricating Intraoral pH Sensor for Continuous Health Monitoring.
Yijing Jiang, Junzhe Jin, **Yunhui Song**, Haiyang Xu, and Michael Wessely. 2025. In Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25).

Teaching Experience

Engineering Interactive Technologies, Aarhus University *Sept. 2025 – Jan. 2026*
Teaching Assistant

Projects

DynamicPanoSynthVR: Dynamic Panoramic VR Experience. *Sept. 2025 – Jan. 2026*
Supervised by Prof. Stefanie Zollmann

- Built a hybrid rendering pipeline that integrates a reconstructed 4D dynamic foreground with a static panoramic background for free-viewpoint VR observation.

X-Block: Making Hybrid Interaction Felt, Seen, and Built. *Sept. 2025 – Dec. 2025*
Supervised by Prof. Eve Hoggn

- Designed a hybrid LEGO-based social deduction icebreaker using an ESP32 wristband, vibrotactile cues, RGB feedback, and tangible models.

Let's Ceramic: An Immersive Journey into Ceramic Heritage and Arts. *Apr. 2025 – Jun. 2025*
Supervised by Prof. Jens Emil Grønbæk

- Designed an AR ceramic-making experience with gesture-driven shaping, decorating, firing, and reflection stages.

VisionNav: Intelligent Vibration-Guided Navigation Companion. *Oct. 2024 – Dec. 2024*
Supervised by Prof. Niels Olof Bouvin

- Built a tactile navigation concept using dual wrist-worn vibration modules for direction and obstacle awareness.

Awards and Honors

Stibo Foundation Travel Grant (DKK 14,000) *May 2025*
Supported attendance at ACM CHI 2025

Outstanding Graduate of Chengdu University of Technology *Jun. 2021*

Bronze Medal, ICPC Asia Regional Contest (Yinchuan Site) *Oct. 2019*

Working Experiences

Huawei Technologies Co., Ltd & FESCO Adecco China
C++ Software Development Engineer *Apr. 2022 – Aug. 2024*